Lecture 15: Splitting Fields and Normal Extensions

MAT205 Abstract Algebra II

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Part I: Splitting Fields

Definition: Splitting Field

Splits. $f \in F[x]$ of degree $n \geq 1$ *splits* in an extension E/F if it factors into linear factors in $E[x]$:

$$f(x) = c\,(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \qquad c \in F^\times,\ \alpha_i \in E.$$

Definition. E/F is a *splitting field for f over F* if

(i) f splits in E, and

(ii) $E = F(\alpha_1, \ldots, \alpha_n)$ — generated over F by the roots of f.

Condition (ii) forces minimality. So $\mathbb{C}$ is *not* a splitting field of x^2-2 over $\mathbb{Q}$: it splits there, but $\mathbb{C} \neq \mathbb{Q}(\sqrt{2})$.

Examples. x^2-2 over $\mathbb{Q}$: $\mathbb{Q}(\sqrt{2})$. x^2+1 over $\mathbb{R}$: $\mathbb{C}$. x^2+x+1 over $\mathbb{F}_2$: $\mathbb{F}_4$.

Existence of Splitting Fields

Convention. Fix an algebraic closure $\overline{F}$ of F. By the isomorphism extension theorem (applied to id_F), every algebraic extension of F embeds into $\overline{F}$, so we view all of them as subfields of $\overline{F}$.

Theorem. Every $f \in F[x]$ of degree $n \geq 1$ has a splitting field E over F inside $\overline{F}$, with $[E : F] \leq n!$.

Proof.

- $\overline{F}$ is algebraically closed, so f has roots $\alpha_1, \ldots, \alpha_n \in \overline{F}$. Put $E = F(\alpha_1, \ldots, \alpha_n)$ — a splitting field.
- Adjoin the roots one at a time: $[F(\alpha_1) : F] \leq n$, then $[F(\alpha_1, \alpha_2) : F(\alpha_1)] \leq n - 1$ (the minimal polynomial of α_2 divides $f/(x - \alpha_1)$), and so on.
- Multiplying, $[E : F] \leq n(n-1)\cdots 1 = n!$. $\square$

Without a closure in hand, the roots can instead be built by the Kronecker step $F \rightsquigarrow F[x]/(p)$, iterated — exactly how $\overline{F}$ itself is constructed.

Uniqueness of Splitting Fields

Inside the fixed $\overline{F}$, the splitting field of f is the one subfield generated by its roots, $E = F(\alpha_1, \ldots, \alpha_n)$ — so "the" splitting field is well-defined. The only question is whether a different algebraic closure gives an isomorphic field.

Theorem (Fraleigh, Exercise 50.25). Any two splitting fields of f over F are isomorphic by an isomorphism fixing F.

Proof.

- Let $E \subseteq \overline{F}$ and $E' \subseteq \overline{F'}$ be splitting fields of f in two algebraic closures of F.
- The two closures are F-isomorphic via some $\lambda : \overline{F} \to \overline{F'}$ (isomorphism extension theorem).
- λ fixes F, so it permutes the roots of f; hence $\lambda(E) = F(\lambda\alpha_1, \ldots, \lambda\alpha_n) = E'$.
- So λ restricts to an isomorphism $E \to E'$. $\square$

Examples of Splitting Fields

- $x^3 - 2$ **over** $\mathbb{Q}$**.** Roots $\sqrt[3]{2},\ \omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$ with $\omega = e^{2\pi i/3}$. Splitting field $\mathbb{Q}(\sqrt[3]{2}, \omega)$:
$$[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = \underbrace{2}_{\omega} \cdot \underbrace{3}_{\sqrt[3]{2}} = 6 = 3!.$$
- $x^4 - 2$ **over** $\mathbb{Q}$**.** Roots $\pm\sqrt[4]{2},\ \pm i\sqrt[4]{2}$. Splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$, degree 8.
- $x^p - 1$ **over** $\mathbb{Q}$ **(**p **prime).** With $\zeta = e^{2\pi i/p}$ a primitive p-th root of unity, the splitting field is $\mathbb{Q}(\zeta)$ (cyclotomic field), of degree $p - 1$.
- $x^{p^n} - x$ **over** $\mathbb{F}_p$**.** Its p^n distinct roots form the field $\mathbb{F}_{p^n}$ — so $\mathbb{F}_{p^n}$ is the splitting field of $x^{p^n} - x$.

Part II: Normal Extensions

Definition: Normal Extension

Definition. An algebraic extension E/F is *normal* if every irreducible $p \in F[x]$ with at least one root in E splits completely in E. Equivalently: E contains all conjugates of each of its elements.

Lemma. Every degree-2 extension is normal. (The second root of a quadratic is $-b/a$ minus the first, so it lies in E too.)

Non-example. $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* normal: $x^3 - 2$ has the root $\sqrt[3]{2}$ but its other roots are non-real, so they are not in $\mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$.

Terminology. Fraleigh's "finite normal extension" (Def 53.1) means a *separable* splitting field — our *Galois* extension. His name for the property above is just "splitting field over F" (Cor 50.6). We keep the modern split.

Examples and Non-examples of Normal Extensions

Check by the characterization: **normal** $\iff$ **splitting field**.

Normal.

- $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ — splitting field of $(x^2-2)(x^2-3)$.
- $\mathbb{C}/\mathbb{R}$ — splitting field of x^2+1.
- $\mathbb{F}_{p^n}/\mathbb{F}_p$ — splitting field of $x^{p^n} - x$.
- $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ — splitting field of $x^n - 1$ (the n-th roots of unity are the powers of ζ_n).
- any degree-2 extension; any algebraic closure $\overline{F}/F$.

Not normal.

- $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ — $x^3 - 2$ has $\sqrt[3]{2}$ but not $\omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$.
- $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ — $x^4 - 2$ has $\sqrt[4]{2}$ but not $\pm i\sqrt[4]{2}$.

Characterization of Finite Normal Extensions

Theorem (Fraleigh 50.3 + Cor 50.6). For a finite extension E/F inside $\overline{F}$, the following are equivalent.

(1) E/F is normal.

(2) E is the splitting field over F of some $f \in F[x]$.

(3) Every F-embedding $\tau : E \to \overline{F}$ has image $\tau(E) = E$ — i.e. every such τ is an automorphism of E.

Fraleigh states (3) as: every automorphism of $\overline{F}$ fixing F maps E onto itself. This is the operational form — a normal extension has no "outward" embeddings, only self-maps.

Characterization: Proof

$(1) \Rightarrow (2)$. Write $E = F(\alpha_1, \ldots, \alpha_r)$ and set $f = \prod_i m_{\alpha_i,F}$. Each $m_{\alpha_i,F}$ has a root in E, so by normality splits in E; thus E is the splitting field of f.

$(2) \Rightarrow (3)$. An F-embedding τ sends each root of f to a root of f (it fixes the coefficients), so it permutes the finite root set. Since E is generated by those roots, $\tau(E) = E$.

$(3) \Rightarrow (1)$. Let $p \in F[x]$ be irreducible with a root $\alpha \in E$, and $\beta \in \overline{F}$ any root of p. The conjugation isomorphism $F(\alpha) \to F(\beta)$ extends to an embedding $\tau : E \to \overline{F}$ with $\tau(\alpha) = \beta$. By (3), $\beta = \tau(\alpha) \in E$. So p splits in E. $\square$

Normal Closure

Theorem. Every finite $E = F(\alpha_1, \dots, \alpha_r)$ sits inside a smallest normal extension of F: the splitting field N of $f = m_{\alpha_1,F} \cdots m_{\alpha_r,F}$, the *normal closure* of E/F.

- N is normal (a splitting field) and contains E; any normal extension containing E contains every conjugate of each α_i, hence contains N — so N is minimal.
- Equivalently, N is the compositum of all images $\tau(E)$, τ ranging over the F-embeddings $E \to \overline{F}$.

Example. $E = \mathbb{Q}(\sqrt[3]{2})$ has three $\mathbb{Q}$-embeddings into $\overline{\mathbb{Q}}$, sending $\sqrt[3]{2}$ to $\sqrt[3]{2},\ \omega\sqrt[3]{2},\ \omega^2\sqrt[3]{2}$. The compositum of their images is $N = \mathbb{Q}(\sqrt[3]{2}, \omega)$, the splitting field of $x^3 - 2$ — degree 6.

Normality under Composites and Intersections

Theorem. If E_1, E_2 are finite normal extensions of F inside $\overline{F}$, then the compositum E_1E_2 and the intersection $E_1 \cap E_2$ are normal over F.

Proof. Use the automorphism form: K/F is normal iff every F-automorphism σ of $\overline{F}$ fixes K setwise. For such σ, normality of E_1, E_2 gives $\sigma(E_1) = E_1$ and $\sigma(E_2) = E_2$, so

$$\sigma(E_1E_2) = E_1E_2, \qquad \sigma(E_1 \cap E_2) = E_1 \cap E_2.$$

Both are stable under every such σ, hence normal. $\square$

By contrast, stacking two normal steps need not give a normal extension — normality is not transitive.

Example. $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ are normal; their compositum $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is normal, their intersection is $\mathbb{Q}$.

Normality Is Not Transitive

Consider the tower

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2}).$$

- Each step has degree 2, hence is normal.
- But $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is *not* normal: $x^4 - 2$ has the root $\sqrt[4]{2}$ but its roots $\pm i\sqrt[4]{2}$ are non-real, so not in $\mathbb{Q}(\sqrt[4]{2}) \subset \mathbb{R}$.

So two stacked normal steps need not give a normal extension. (By contrast, algebraicity is transitive.) What is always true: if E/F is normal, then E/K is normal for every intermediate K.

The Index $\{E : F\}$ and the Bound $\{E : F\} \leq [E : F]$

Definition. $\{E : F\}$ is the number of F-embeddings $E \to \overline{F}$.

The chain.

$$|\mathrm{Aut}(E/F)| \ \leq \ \{E : F\} \ \leq \ [E : F].$$

- $|\mathrm{Aut}(E/F)| \leq \{E : F\}$: every F-automorphism of E is in particular an F-embedding $E \to \overline{F}$ (its image is E).
- $\{E : F\} \leq [E : F]$: for $E = F(\alpha)$, an F-embedding sends α to a root of m_α, and m_α has at most $\deg m_\alpha = [F(\alpha) : F]$ roots, so $\{F(\alpha) : F\} \leq [F(\alpha) : F]$. Index and degree are both multiplicative over a tower, so the bound multiplies up to $\{E : F\} \leq [E : F]$.

Index and Normality

Two equalities in the chain $|\mathrm{Aut}(E/F)| \leq \{E : F\} \leq [E : F]$, two properties:

- $|\mathrm{Aut}(E/F)| = \{E : F\} \iff E/F$ normal (Fraleigh Cor 50.7): the embeddings with image E are exactly the automorphisms, so the counts agree iff *every* embedding has image E — which is normality.
- $\{E : F\} = [E : F] \iff E/F$ separable: the bound $\{F(\alpha) : F\} \leq [F(\alpha) : F]$ is an equality iff m_α has $\deg m_\alpha$ *distinct* roots, i.e. no repeated roots — the definition of α separable; multiplicativity lifts this to E/F.

Separable regime. Over a perfect field (char 0 or finite — most of this course), $\{E : F\} = [E : F]$ automatically, so $|\mathrm{Aut}(E/F)| = [E : F] \iff E/F$ normal.

Galois Extensions = Normal + Separable

Theorem. For a finite extension E/F,

$$|\mathrm{Aut}(E/F)| = [E : F] \quad \iff \quad E/F \text{ normal and separable.}$$

Both inequalities $|\mathrm{Aut}(E/F)| \leq \{E : F\} \leq [E : F]$ become equalities: normal closes the left, separable the right.

Definition. Such an E/F is a *Galois extension*; one writes $\mathrm{Gal}(E/F) := \mathrm{Aut}(E/F)$. (This is Fraleigh's "finite normal extension", Def 53.1.)

- $\mathbb{F}_{p^n}/\mathbb{F}_p$: Galois, $\mathrm{Gal} \cong \mathbb{Z}/n\mathbb{Z}$.
- $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$: Galois, $|\mathrm{Gal}| = 4$.
- $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$: not Galois — $|\mathrm{Aut}| = 1 < 3$, the deficit is non-normality.

Worked Example: $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ Is Galois

$E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $[E : \mathbb{Q}] = 4$, basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$. A $\mathbb{Q}$-automorphism sends $\sqrt{2} \mapsto \pm\sqrt{2}$ and $\sqrt{3} \mapsto \pm\sqrt{3}$ — four of them:

	id	a	b	c
$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$	$-\sqrt{2}$	$-\sqrt{2}$
$\sqrt{3}$	$\sqrt{3}$	$-\sqrt{3}$	$\sqrt{3}$	$-\sqrt{3}$

- $\mathrm{Gal}(E/\mathbb{Q}) = \{\mathrm{id}, a, b, c\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, with $|\mathrm{Gal}| = 4 = [E : \mathbb{Q}]$.
- E is the splitting field of $(x^2 - 2)(x^2 - 3)$ (normal) and separable — hence Galois.
- Fixed fields: $\langle a \rangle \to \mathbb{Q}(\sqrt{2})$, $\langle b \rangle \to \mathbb{Q}(\sqrt{3})$, $\langle c \rangle \to \mathbb{Q}(\sqrt{6})$ — three subgroups matching three intermediate fields, the Galois correspondence.